

Gait Phenotyping Across Pathologies Using Wearable IMUs: Temporal Asymmetry, Sensor Optimization, and Clinical Screening

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Abstract

Wearable inertial measurement units (IMUs) enable scalable, real-world gait analysis for diverse clinical populations. This study makes two primary contributions: (1) characterizing stride and step temporal asymmetry across neurological and orthopaedic subtypes using the multi-pathology dataset of Voisard et al. (2025), and (2) evaluating IMU sensor placement trade-offs for gait-based pathology classification. A parameter extraction pipeline computing seven clinically validated temporal metrics was developed, and a sensor ablation study was conducted using SVM, XGBoost, and Random Forest models. Results suggest that healthy subjects exhibit greater temporal asymmetry than pathological groups (Cohen's $d = 0.77$, 95% CI: 0.50–1.07, $p < 0.001$), consistent with a motor lateralization hypothesis. Across all models, the right foot (RF) sensor consistently provided the highest single-sensor performance (up to 88.9% balanced accuracy), and foot-only configurations retained approximately 93% of full-sensor accuracy—suggesting potential for a practical two-sensor clinical wearable. These findings indicate a possible role for IMU-derived temporal asymmetry as a clinical indicator, with further validation needed before clinical translation.

I. INTRODUCTION

Human walking is among the most precisely regulated motor behaviors executed by the central nervous system. Its bilateral coordination reflects the integrity of a multilevel neuromotor architecture, and its degradation is a sensitive indicator of diverse neurological and orthopaedic conditions [1, 2].

Quantitative gait analysis has long been a clinical tool, but its reach has historically been limited by the cost and spatial constraints of laboratory-based force-platform and motion-capture systems. Inertial measurement units (IMUs) have disrupted this paradigm: lightweight, wireless, and low-cost, they enable gait measurement in clinics, rehabilitation wards, and home environments. The dataset published by Voisard et al. (2025) in Scientific Data provides 1,356 trials from 260 participants across three clinical categories and seven diagnostic subtypes, recorded with four synchronised IMUs [3].

This paper reports two contributions: (1) statistical characterization of temporal asymmetry across all seven diagnostic subtypes, and (2) sensor ablation analysis identifying cost-effective IMU configurations. We frame our findings as preliminary evidence warranting further validation rather than established clinical guidelines.

II. RESEARCH QUESTIONS

RQ1 (Sensor Optimization): Which sensor configurations provide optimal cost-performance trade-offs for pathology classification across gait phases?

RQ2 (Temporal Asymmetry): Does temporal gait asymmetry differ between healthy and pathological populations, and if so, in which direction?

III. MAGNITUDE OF THE PROBLEM

Gait disorders affect hundreds of millions of people worldwide. Neurological conditions alone affect more than one billion people globally, with motor and coordination deficits that directly impair walking function [4]. Parkinson's disease affects approximately 8.5 million people worldwide [5]. Stroke affects ~13 million people annually, with 30–40% of survivors exhibiting persistent gait abnormalities [6]. On the orthopaedic side, knee osteoarthritis affects ~250 million people globally and is the leading cause of pain-related mobility impairment in adults over 50 [8].

Falls are the leading cause of injury-related mortality among adults aged 65 and older. The CDC estimates ~36 million falls annually among older adults in the U.S., resulting in over 32,000 deaths and costs exceeding \$50 billion per year [10]. Objective gait monitoring may support earlier detection and intervention.

IV. RELEVANCE

Wearable IMU-based gait analysis potentially addresses three clinical needs: screening, longitudinal monitoring, and rehabilitation assessment. The Voisard et al. (2025) dataset enables direct validation of computational gait features against established clinical measures including UPDRS-III, mRS, FAC, TNS, Kellgren–Lawrence grading, and Visual Gait Assessment (VGA).

V. WHAT HAS BEEN DONE WITH THE DATASET

The dataset creators validated automatic gait event detection from healthy to severely impaired patients [12]. Al-Harhi et al. (2026) applied G-MASA-TCN deep learning to all 1,356 trials, with Integrated Gradients explainability identifying foot-mounted IMU channels as the most attribution-weighted inputs—converging with our sensor ablation finding that RF is the single most informative placement [3].

Sadeghsalehi (2025) applied a multi-stream attention model to binary classification tasks and identified a critical laterality confound: all 15 HOA patients had right-sided pathology, and 47 of 49 CVA patients had right-dominant motor deficits. This means right-foot sensor dominance in those tasks reflects dataset-specific laterality bias rather than a general superiority of RF sensing [11]. We flag this where relevant in our subtype results.

VI. METHODOLOGY

a. Data Preparation

All 1,356 trials from 260 subjects were downloaded from Figshare (DOI: 10.6084/m9.figshare.28806086). U-turn segments were excluded from parameter computations as they represent biomechanically distinct, non-steady-state gait (heel-strikes outside [uturn_start, uturn_end] were retained). After exclusion, 974 valid straight-walking trials from 216 subjects were retained.

b. Parameter Extraction

Seven clinically validated gait metrics were computed from pre-annotated bilateral heel-strike timestamps per trial: stride time, step time, cadence, stance phase, swing phase, stride CV, and symmetry index. All parameters were aggregated at the subject level (mean across retained trials).

c. Sensor Ablation

A sensor ablation study was performed using XGBoost, SVM, and Random Forest models across gait phases (full gait, pre-uturn, post-uturn, u-turn), window sizes (1–6 s), and overlap percentages (0%, 25%, 50%). Performance was assessed using balanced accuracy and macro-F1 to account for class imbalance.

VII. PRELIMINARY RESULTS

a. Gait Parameters Across Groups

Neurological groups (PD, CVA, RIL) exhibit elevated stride times, decreased cadence, increased temporal variability (CV), and reduced symmetry index compared to healthy subjects. Orthopaedic groups generally show intermediate or near-healthy values, with ACL patients indistinguishable from healthy controls on most temporal parameters, consistent with an intact central motor program driving a mechanically altered but neurally intact gait.

TABLE I. SUBJECT-LEVEL TEMPORAL ASYMMETRY METRICS. 95% Bootstrap CI shown for primary statistic. *** $p < 0.001$, ** $p < 0.01$ (Welch's t -test, Healthy vs. Combined Pathological).

Metric	Healthy (n=70)	Ortho (n=35)	Neuro (n=111)
Stride AI	0.052 ± 0.036	0.039 ± 0.027	0.029 ± 0.018
Stride L-R (s)	0.069 ± 0.054	0.054 ± 0.037	0.038 ± 0.023
Step AI	0.148 ± 0.091	0.088 ± 0.075	0.118 ± 0.093
Step CV (L)	0.284 ± 0.236	0.176 ± 0.153	0.187 ± 0.152
Mean step time (s)	0.605 ± 0.035	0.628 ± 0.053	0.608 ± 0.057

b. Temporal Asymmetry

(1) Healthy vs. Pathological. The central finding is that healthy subjects exhibit significantly larger temporal asymmetry than pathological groups (Table I). Stride |AI| yields Cohen's $d = 0.77$ (95% Bootstrap CI: 0.50–1.07) with $p < 0.001$. AUC = 0.716 (95% CI: 0.635–0.792) at threshold 0.049 delivers 83% specificity and 59% sensitivity. Mean step time does not differ between groups ($p = 0.19$), ruling out walking speed as a confound.

In healthy individuals, the signed AI is consistently positive (left stride slightly longer, mean = +0.009), consistent with dominant-leg lateralization. In neurological patients it approaches zero and becomes directionally unpredictable. The orthopaedic group is not significantly different from healthy (LME $p = 1.00$), suggesting mechanical joint pathology does not disrupt central lateralization.

(2) Cross-Pathology Comparison. Cohen's d (signed: pathological – healthy) indicates that neurological conditions with central motor involvement show the largest asymmetry reduction (Table II). RIL ($d = -0.87$) and PD ($d = -0.77$) show large effects; CVA ($d = -0.73$) shows a large effect, though interpretation must account for the right-dominant laterality confound [11]. CIPN shows a medium effect ($d = -0.53$). KOA reaches statistical significance ($d = -0.45$, $p = 0.034$), while HOA and ACL are not significantly different from healthy.

TABLE II. STRIDE |AI| BY SUBTYPE (REVISED). Cohen's $d = (\text{pathological} - \text{healthy}) / \text{pooled SD}$; negative values indicate lower |AI| than healthy. †CVA and HOA subject to right-dominant laterality sampling confound; interpret with caution. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Subtype	Category	Stride AI	d vs Healthy	p-value
RIL (n=14)	Neurological	0.024	-0.87	<0.001***
PD (n=17)	Neurological	0.025	-0.77	<0.001***
CVA (n=44)†	Neurological	0.027	-0.73	<0.001***
CIPN (n=36)	Neurological	0.033	-0.53	0.003**
KOA (n=14)	Orthopaedic	0.036	-0.45	0.034*
HOA (n=12)†	Orthopaedic	0.042	-0.27	0.226 ns
ACL (n=9)	Orthopaedic	0.049	-0.09	0.779 ns

(3) ROC Screening Performance and Clinical Validation. Stride |AI| achieves AUC = 0.716 [0.635–0.792] at threshold 0.049, sensitivity 0.59, specificity 0.83. Spearman correlation between VGA severity and stride |AI| is $\rho = -0.206$ ($p < 0.001$, $n = 927$), indicating convergent validity with established clinical assessment. This correlation is weak-to-moderate ($r^2 \approx 0.042$), suggesting IMU captures information beyond coarse visual rating but that the relationship is not strong.

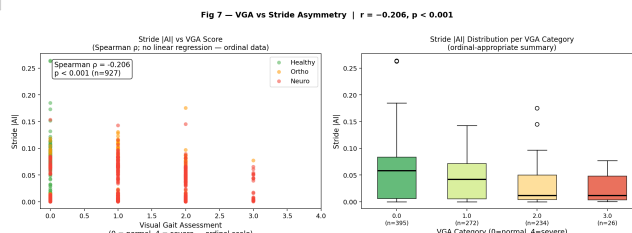


Fig. 7: VGA score vs stride |AI|. Panel A: scatter colored by group; linear regression not shown (VGA is an ordinal variable). Spearman $\rho = -0.206$, $p < 0.001$, $n = 927$. Panel B: stride |AI| distribution per VGA category (0 = normal, 4 = severe).

c. ML Feature Integration

Adding 11 asymmetry features to the 217-feature sensor set produced a modest but consistent improvement (Table III). On the matched 974-trial subset, macro-F1 improved +0.007 and

BACC +0.009. Asymmetry features alone achieve ~50% F1 (vs. 33% chance). No asymmetry feature appears in the RF top-25, confirming spectral features already implicitly encode temporal lateralization. The AUC of 0.716 represents a modest result; the primary contribution of this work is the *clinical characterization* of asymmetry patterns across subtypes, not incremental model performance improvement.

TABLE III. 5-FOLD STRATIFIEDGROUPKFOLD CV (SUBJECT-GROUPED). *Values after slash = matched 974-trial subset.

Configuration	Features	LR F1	RF F1	SVM F1
Sensor only	217	0.748	0.810	0.816 / 0.806*
Sensor + Asymmetry	228	0.741	0.810	0.822 / 0.812*
Feet + Asymmetry	120	0.735	0.770	0.791
Asymmetry only	11	0.447	0.466	0.491

d. Feature Directionality

Of 182 features significant at $p < 0.05$ (Kruskal-Wallis), the dominant pattern is monotonic decrease H>O>N (67 features). Spectral centroid frequencies and signal dynamics decrease systematically from healthy to neurological, reflecting slower and less variable gait. All top-20 features show 93–100% cross-subject concordance, confirming effects are present in nearly every individual.

e. Sensor Ablation Results

Key Finding: Foot-only sensors (LF+RF) retain approximately 93% of full-sensor balanced accuracy (68.2% vs 73.4%), suggesting a practical two-sensor wearable configuration for clinical use. The RF sensor is consistently the top single-sensor recommendation across all three models and gait phases.

(1) SVM. The RF sensor achieved the highest single-sensor SVM performance. Best results: full gait 85.97% BACC / 85.4% F1 (5s, 50% overlap); pre-urn 83.2% / 80.8% (4s, 25%); post-urn 87.6% / 80.8% (6s, 25%). Best two-sensor configuration: LB+HE for post-urn (89.8% / 84.4%, 6s, 50%).

TABLE IV. BEST SENSOR CONFIGURATIONS — SVM.

Phase	Best Single	Win	OL%	BACC	F1
Full gait	RF	5s	50%	85.97%	85.4%
Pre-urn	RF	4s	25%	83.2%	80.8%
Post-urn	RF	6s	25%	87.6%	80.8%
U-turn	RF	1.28s	50%	76.0%	75.3%

(2) XGBoost. Best single-sensor: full gait RF 88.7% (5s, 50%); post-urn LB 90.7% (3s, 25%); pre-urn HE 90.0% (6s, 50%); u-turn RF 78.2% (1s, 50%). Best two-sensor: post-urn HE+LB 93.9% (6s, 25%); full gait LB+RF 91.8% (5s, 50%).

(3) Random Forest. Best single-sensor: pre-urn RF 88.9% / 85.6% (3s, 25%); post-urn RF 88.5% / 82.0% (6s, 50%); full gait RF 85.1% / 84.2% (5s, 50%). Best two-sensor: post-urn LF+RF 89.1%, full gait LB+HE 87.0%.

Window size and overlap significantly impact performance. Larger windows (5–6 s) are generally optimal for continuous gait phases across models, while the u-turn phase requires smaller windows (1–1.28 s) to capture its biomechanical signature.

VIII. DISCUSSION

This study makes two contributions. First, we characterize temporal asymmetry across seven gait pathology subtypes, finding that healthy subjects exhibit greater unsigned asymmetry than pathological groups ($d = 0.77$, $p < 0.001$). This is consistent with Sadeghi et al.'s meta-analysis showing reliable left-right differences in able-bodied adults driven by motor lateralization [14]. Our data suggest that neurological conditions disrupt the lateralized motor program, causing convergence toward zero asymmetry. Notably, RIL and PD show the largest effects ($d = -0.87$ and -0.77), while orthopaedic subtypes show smaller or non-significant effects, consistent with preserved central motor control in mechanical joint conditions.

Second, the sensor ablation study indicates that the right foot sensor is consistently the most informative single placement. Foot-only configurations retain approximately 93% of full-sensor accuracy, suggesting potential for a practical two-sensor wearable. However, the laterality confound in CVA and HOA (Sadeghsalehi et al.) means that RF dominance in those tasks may partly reflect dataset structure rather than biological generalizability [11].

Clinical Screening: AUC = 0.716 at threshold 0.049 provides 83% specificity, meaning negative screens reliably identify healthy individuals. Sensitivity is modest (59%), and the AUC itself represents a modest classification result. The ML improvement from adding asymmetry features (+0.007 F1) is small and should not be over-interpreted. The primary value of this analysis is the characterization of asymmetry patterns across pathology subtypes, not a state-of-the-art classification system.

The weak-to-moderate VGA–IMU correlation ($\rho = -0.206$, $r^2 \approx 0.042$) indicates that IMU-derived temporal asymmetry captures information beyond coarse visual assessment, but that VGA alone explains only ~4% of the variance in stride |AI|. Future work should examine whether combining VGA with IMU metrics improves clinical utility.

IX. LIMITATIONS

The dataset has several important limitations. (1) The right-dominant laterality confound in CVA (47/49) and HOA (15/15) means that asymmetry directionality results for these subtypes are not generalizable. (2) Small subtype sample sizes (ACL $n=9$, RIL $n=14$, KOA $n=14$) limit statistical power for per-subtype comparisons; bootstrap CIs reflect this uncertainty. (3) Results are from a single cohort in a controlled 10-metre walk protocol; ecological validity in real-world settings is unknown. (4) The ML models use 5-fold StratifiedGroupKFold with subject grouping to prevent data leakage, but replication in an independent dataset is necessary before any clinical claims.

X. CONCLUSION

We analyzed temporal gait asymmetry and sensor placement efficiency in a multi-pathology IMU dataset. Our results suggest that: (1) healthy subjects exhibit greater temporal asymmetry than pathological groups, consistent with a motor lateralization hypothesis; (2) neurological subtypes (RIL, PD, CVA) show the largest asymmetry reductions; and (3) foot-only sensors may offer a practical two-sensor clinical configuration. These findings indicate potential clinical utility of IMU-derived temporal asymmetry as a gait indicator, subject to validation in larger independent cohorts and real-world settings.

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